

AI PLACEMENT PREPARATION AGENT

An Automated Agentic AI Assistant using n8n, Docker, Google Sheets, & Gemini API

Course:	B.Tech (CSE - AI)
Project Type:	Individual Mini Project
Submission Date:	August 10, 2026
Technologies Used:	Docker, n8n, Google Gemini API, Google Sheets, Gmail API

1. Project Objective

The objective of this project is to construct an AI-powered **Placement Preparation Assistant** that automatically delivers customized resources (aptitude questions, coding problems, interview tips, and resume reviews) to students preparing for campus placements. By leveraging workflow automation platforms (n8n) and generative AI model endpoints (Gemini API), the system replaces manual search routines with automated daily content curation and analysis.

This project provides practical, hands-on experience in configuring **Agentic AI Workflows**, managing local software environments via **Docker**, and securely integrating external service APIs (Google Workspace & Gemini).

2. System Architecture & Workflows

The assistant is structured as a containerized flow running locally via Docker. The workflows are defined inside n8n, which connects directly to Google Sheets (serving as the database of student information), calls Google AI Studio's Gemini models via secure HTTP endpoints, and communicates with students via Gmail.

Daily Prep Workflow Step-by-Step Execution:

- **Cron Node:** Triggers every morning at 9:00 AM.
- **Google Sheets Reader:** Queries the student registry sheet, pulling subscriber lists containing 'Name' and 'Email'.
- **Loop Over Items:** Evaluates each student record to perform personalized generation.
- **HTTP Request (Gemini API):** Generates unique, freshman-level aptitude questions, coding problems, and interview tips on the fly.
- **Merge Node:** Aggregates JSON structures returned from each individual Gemini prompt.
- **Google Sheets Writer:** Automatically saves today's generated question details alongside the timestamp for future audit.
- **Gmail Service:** Composes a highly formatted HTML mail containing the questions, explanations, motivational quotes, and fires it directly to the student's inbox.

3. Resume Feedback Workflow

In addition to the daily prep scheduler, a dedicated event-driven workflow processes resume uploads. When a student uploads a Resume.pdf to Google Drive, the file is monitored by n8n. The system extracts the textual contents of the PDF, sends the raw text along with structured analysis guidelines to the Gemini API, and returns actionable resume reviews including parsed skills, highlights of weaknesses, strategic advice, and an estimated ATS alignment score directly to the student.

4. Technical Matrix

Technology	Specific Purpose
Docker	Containerizes the local runtime, ensuring instant, reproducible deployments of n8n server environments.
n8n	Coordinates complex conditional pipelines, triggers, loops, API authentications, and data mapping.
Gemini API	Utilizes gemini-1.5-flash for high-speed, cost-efficient, context-aware content generation.
Google Sheets	Serves as the low-code student relational database storing student names, emails, and historic questions.
Gmail API	Ensures robust deliverability of placement prep packages directly to student accounts.

5. Summary of Outcomes

By automating these placement tasks, B.Tech students achieve robust familiarity with **Agentic AI pipelines** and infrastructure orchestration. This automation saves time, drives student engagement with consistent mock preparation, and provides an end-to-end blueprint for building scalable, real-world productivity agents.